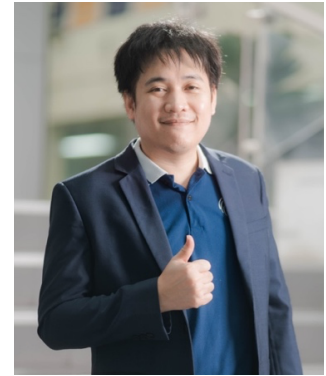


WORAWIT TEPSAN, Ph.D.

Current Work: Postdoctoral Research Fellow under the British Council ISPF Early Career Fellowship Programme, Edinburgh Napier University, United Kingdom (April 2026–March 2027)

Permanent Position: Lecturer, International College of Digital Innovation (ICDI), Chiang Mai University, Chiang Mai, Thailand (2020–Present)

Research Interests: Data Analytics, Machine Learning, Artificial Intelligence, Applied Mathematics, and Applications in Digital Health.



Education:

- Ph.D. in Mathematics, University of Houston, USA, 2020
- M.Sc. in Mathematics, Chulalongkorn University, Thailand, 2013
- B.Sc. in Mathematics, with a Minor in Computer Science, Chiang Mai University, Thailand, 2010

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Public Research Profiles:

- ORCID: <https://orcid.org/0000-0001-6423-9633>
- Google Scholar: <https://scholar.google.com/citations?user=1QJ3-AcAAAAJ&hl=en>
- Scopus: <https://www.scopus.com/authid/detail.uri?authorId=57219787378>
- GitHub: <https://github.com/wtepsan>

Research Metrics (Accessed 13 July 2026):

- Scopus Documents: 15
- Scopus h-index: 5
- Scopus Citations: 362

SELECTED PUBLICATIONS

- P. Phiphatkunarnon, A. Kitro, B. Suksatit, B.-L. Neo, D. Tran, and **W. Tepsan**, “A health informatics framework for integrating machine learning and generative AI in HIV risk stratification and personalized PrEP recommendation,” *Informatics*, vol. 13, no. 7, Art. no. 103, 2026, doi: 10.3390/informatics13070103. 2026
- F. Han and **W. Tepsan**, “Food live streaming e-commerce in China: Enhancing consumer purchase intentions through sensory marketing,” *British Food Journal*, vol. 128, no. 1, pp. 255–273, 2026, doi: 10.1108/BFJ-02-2025-0140. 2026
- F. Han and **W. Tepsan**, “The impact of streamer social presence and professionalism on food live streaming purchase intention: A model of moderated mediation,” *Asia Pacific Journal of Marketing and Logistics*, 2025, doi: 10.1108/APJML-01-2025-0151. 2025
- **W. Tepsan**, A. Phaphuangwittayakul, S. Sokantika, and N. Harnpornchai, “Identifying hubs through influential nodes in transportation networks using a gravity centrality approach,” *Algorithms*, vol. 18, no. 6, Art. no. 356, 2025, doi: 10.3390/a18060356. 2025
- P. Phiphatkunarnon, N. Phanuphak, R. Janamnuysook, D. Tran, B.-L. Neo, and **W. Tepsan**, “Comprehensive analysis and leveraging online innovations to improve HIV and STI prevention and treatment services in major cities of Thailand,” *AIDS and Behavior*, 2025, doi: 10.1007/s10461-025-04714-x. 2025

- P. Lueangwitchajaroen, S. Watcharapinchai, **W. Tepsan**, and S. Sooksatra, “Multi-level feature fusion in CNN-based human action recognition: A case study on EfficientNet-B7,” Journal of Imaging, vol. 10, no. 12, Art. no. 320, 2024, doi: 10.3390/jimaging10120320. 2024
- **W. Tepsan**, S. Sooksatra, P. Lueangwitchajaroen, and S. Watcharapinchai, “Adaptive body part ROI with triple stream approach for human action recognition,” in Proc. 21st Int. Joint Conf. Computer Science and Software Engineering (JCSSE), 2024, pp. 79–85, doi: 10.1109/JCSSE61278.2024.10613745. 2024
- H. Shu, A. Y. Dawod, **W. Tepsan**, L. Mou, and Z. Tang, “Multi-channel microseismic signal classification with convolutional neural networks,” IAES International Journal of Artificial Intelligence, vol. 13, no. 1, pp. 1038–1049, 2024, doi: 10.11591/ijai.v13.i1.pp1038-1049. 2024
- H. Shu, A. Y. Dawod, L. Mu, and **W. Tepsan**, “A survey of machine learning applications in microseismic signal recognition and classification,” in Proc. 15th Int. Conf. Software, Knowledge, Information Management and Applications (SKIMA), Kuala Lumpur, Malaysia, 2023, pp. 18–23, doi: 10.1109/SKIMA59232.2023.10387351. 2023
- X. Ran, X. Zhou, M. Lei, **W. Tepsan**, and W. Deng, “A novel K-means clustering algorithm with a noise algorithm for capturing urban hotspots,” Applied Sciences, vol. 11, no. 23, Art. no. 11202, 2021, doi: 10.3390/app112311202. 2021
- D. P. Blecher and **W. Tepsan**, “Real operator algebras and real positive maps,” Integral Equations and Operator Theory, vol. 93, no. 5, Art. no. 49, 2021, doi: 10.1007/s00020-021-02665-1. 2021

ONGOING RESEARCH PROJECTS

- **AI-Enabled Falls Prevention Devices for Elderly Populations (THER Project, Edinburgh Napier University)** – Developing and evaluating AI-driven wearable sensing and machine learning approaches for monitoring movement, identifying fall-risk indicators, and supporting fall prevention among older adults.
- **Human Action Recognition with Skeleton Databases (NECTEC Collaboration)** – Researching deep learning models to recognize and classify human activities from skeleton-based datasets.
- **AI Agent for Personalized Sexual Health Recommendations** – Developing an AI decision-support tool that provides individualized guidance on HIV prevention (PrEP) and broader sexual health practices tailored to patient risk profiles.
- **Wound Segmentation, Area & Size Measurement** – Building a computer vision pipeline to automatically segment wounds and calculate healing-related metrics for digital health applications.
- **NurseBesties – AI Agent for Nursing Students** – Designing an AI assistant that helps nursing students practice diagnostic reasoning, patient interaction, and clinical decision-making.

ACADEMIC EXPERIENCES

- Coach of Team Optimizer in the Super AI Engineer Program Season 4, 5 and 6 organized by the Artificial Intelligence Association of Thailand (AIAT). 2024 - 2026
- Training in Super AI Engineer Program Season 2 organized by Artificial Intelligence Association of Thailand (AIAT) 2022
- Co-Researcher: The study of growth poles of the Northern Economic Corridor Project, Thailand (Grant from the Ministry of Higher Education, Science, Research and Innovation, Thailand) 2021
- PhD Thesis: Real Operator Spaces, Real Operator Algebras and Real Jordan Operator Algebras 2020
- Master’s degree Thesis: Triangular Mean-Value Functional Equation 2012
- Senior Research: Algorithms for Solving Periodic Pentadiagonal Linear System and Inverting Periodic Pentadiagonal Matrices 2010
- Mini Project: Algorithm for Finding Shortest path in Chiang Mai City 2008

EMPLOYMENT

- Postdoctoral Research Fellow under the British Council ISPF Early Career Fellowship Programme, Edinburgh Napier University, United Kingdom April 2026–March 2027
- Lecturer at International College of Digital Innovation, Chiang Mai University, Thailand Dec 2020 - Current
- Special Expert at International College of Digital Innovation, Chiang Mai University, Thailand Aug 2020 - Dec2020
- Teaching Assistance at University of Houston, TX, USA 2015 - 2020
- Tutor (Self Employed) for secondary and high school students in Math, BKK, Thailand 2010 - 2013

EDUCATION

- PhD in Mathematics from University of Houston, Texas, USA 2015-2020
- Studied for PhD in Mathematics but then moved to University of Houston Ohio University, Ohio, USA (did not graduate) 2013-2015
- Master of Science in Mathematics from Chulalongkorn University, Bangkok, Thailand 2010-2013
- Bachelor of Science in Mathematics (Minor in Computer Science) from Chiang Mai University, Chiang Mai, Thailand 2006-2010

HONORS AND AWARDS

- Abroad fellowship from the Institute for the Promotion of Teaching Science and Technology Project (DPST), Ministry of Education, Thailand 2013-2018
- Scholarship from Development and Promotion of Science and Technology Talents Project (DPST), Ministry of Education, Thailand 2003-2013
- Scholarship from Junior Scientist Talent Project (JSTP), Thailand 2007

TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript
- **Machine Learning and Deep Learning:** PyTorch, model development, training, evaluation, and data analysis
- **Generative AI:** Large language models, AI application development, prompt-based systems, and API integration
- **Backend Development:** FastAPI, RESTful APIs, and server-side application development
- **Deployment and Computing:** Model deployment, cloud-based systems, and high-performance computing